

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 37

WORLD CLIMATIC TYPES

THE STEPPE CLIMATE

- **DISTRIBUTION AND CLIMATE**

TABLE OF CONTENTS

GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Equatorial zone	0-10° N & S	1. Hot, Wet Equatorial climate
Hot Zone	10°-30° N & S	2. a) Tropical Monsoon b) Tropical Marine
		3. Sudan type
		4. Desert type a) Saharan type b) Mid-Latitude type
Warm Temperature Zone	30°-40° N & S	5. Western Margin Mediterranean Type
		6. Central Continental Steppe type.
		7. Eastern Margin a) China type b) Gulf type C) Natal type

COLD ZONE	<u>Arctic or Polar Type</u>			90°
	Tundra vegetation, mosses, lichens			
COOL TEMPERATE ZONE				65° N
	Western Margin <u>British type</u> Deciduous forests	Central Continent <u>Siberian type</u> Evergreen, coniferous forests	Eastern Margin <u>Laurentian type</u> Mixed forests	
WARM TEMPERATE ZONE				45° N
	<u>Mediterranean type</u> Mediterranean Forests & shrubs	<u>Steppe type</u> Steppe temperate grassland	<u>China type</u> Warm wet forests	
HOT ZONE				30° N
	<u>Hot desert</u> Desert vegetation	<u>Sudan type</u> Savanna, tropical grassland	<u>Monsoonal type</u> Monsoon forests	
EQUATORIAL ZONE				10° N
	<u>Hot Wet Equatorial Climate</u> Equatorial rain forests			0°

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GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Cool Temperature Zone	45°-65° N & S	8. Western Margin - British type.
		9. Siberian type - Central Continental
		10. Laurentian type - Eastern Margin
Cold Zone	65°-90° N & S	11. Arctic or Polar or Tundra climate.

PREVIOUS YEARS' PRELIMS QUESTION

A geographic region has the following distinct characteristics:

1. Warm and dry climate
2. Mild and wet winter
3. Evergreen oak trees

The above features are the distinct characteristics of which one of the following regions?

- | | |
|------------------|------------------------------------|
| a) Mediterranean | b) Eastern China |
| c) Central Asia | d) Atlantic coast of North America |

Answer: A

COLD ZONE	<u>Arctic or Polar Type</u>			90°
	Tundra vegetation, mosses, lichens			
COOL TEMPERATE ZONE				65° N
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	<u>Hot Wet Equatorial Climate</u> Equatorial rain forests			0°

PREVIOUS YEARS' PRELIMS QUESTION

Consider the following statements:

1. In equatorial regions, the year is divided into four main seasons.
2. In Mediterranean region, summer receives more rain.
3. In China type climate, rainfall occurs throughout the year.
4. Tropical highlands exhibit vertical zonation of different climates.

Which of the above are correct?

- | | |
|---------------------|------------------|
| a) All of the above | b) 1, 2 & 3 only |
| c) 1, 2 & 4 only | d) 3 & 4 only |

Answer: D

WORLD CLIMATIC TYPES

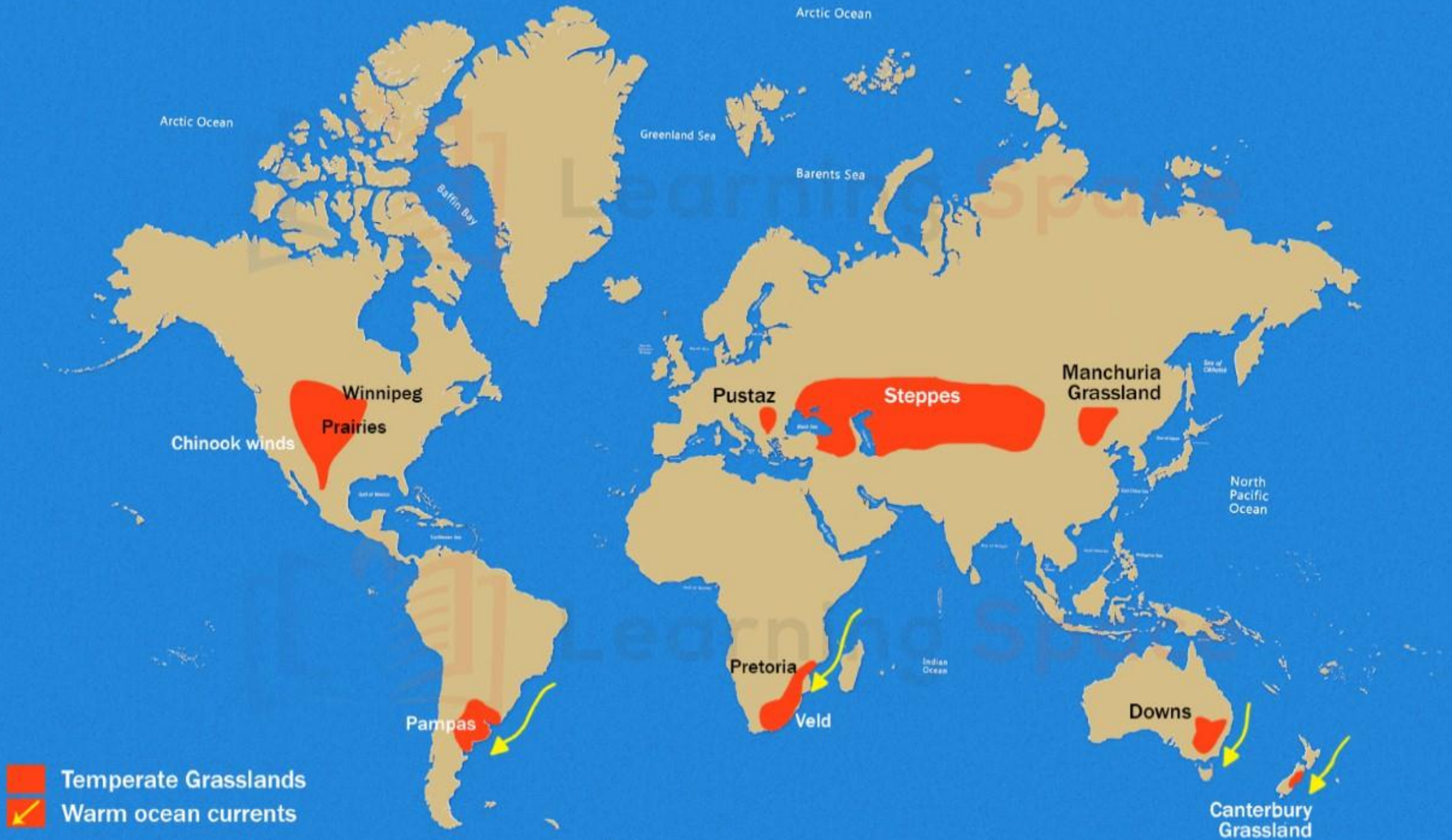
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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

DISTRIBUTION



- 1 Bordering the deserts, away from the Mediterranean regions and in the interiors of continents are the **temperate grasslands**.
- 2 Though they lie in the Westerly wind belt, they are so remote from maritime influence that the grasslands are practically treeless.
- 3 These grasslands are so distinctive in their natural vegetation that, although those which occur in the southern hemisphere have much more moderate climate, they are often dealt with together.



WORLD CLIMATIC TYPES

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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

DISTRIBUTION



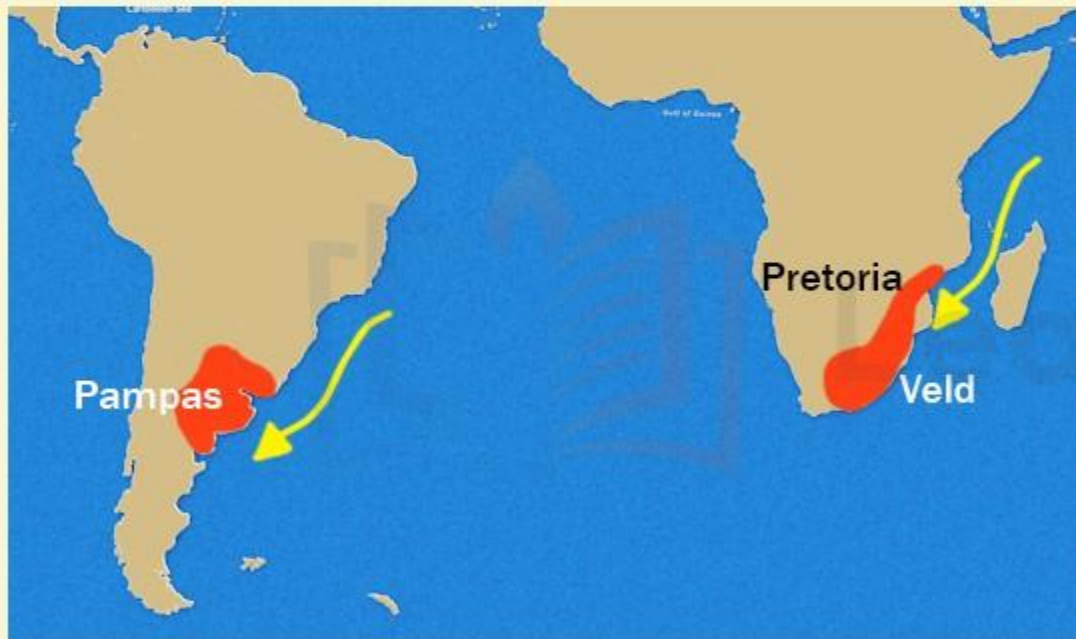
- 4 In the northern hemisphere, the grasslands are far more extensive and are entirely continental. In Eurasia, they are called the Steppes.
- 5 In North America, the grasslands are also quite extensive and are called Prairies.
- 6 They lie between the foothills of the Rockies and the Great lakes.
- 7 In the Southern hemisphere, the grasslands are rather restricted and less continental.

WORLD CLIMATIC TYPES

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

DISTRIBUTION



8

In the case of the **Pampas** of Argentina and Uruguay, the grasslands extend right to the sea and enjoy much maritime influence.

9

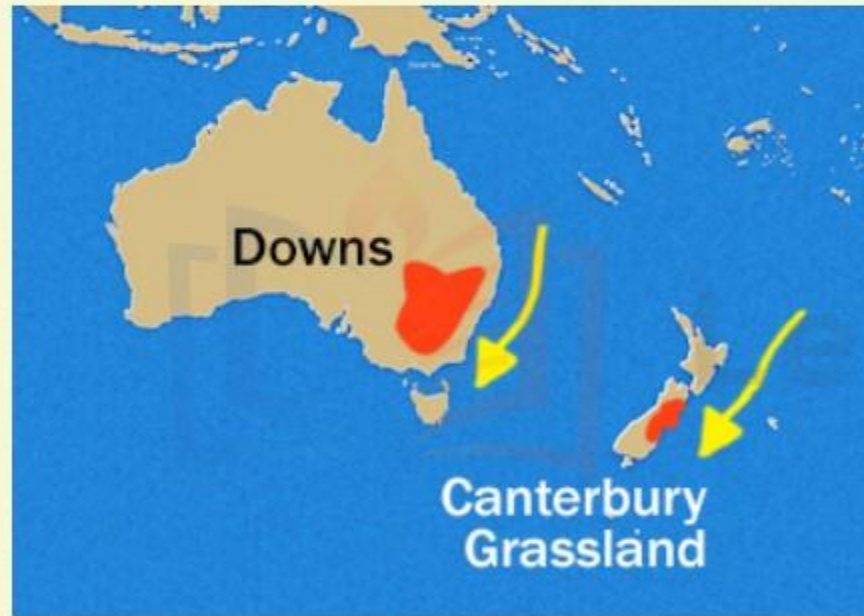
In South Africa, the grasslands are sandwiched between the Drakensberg and the Kalahari Desert and are further sub-divided into the more tropical **Bush-veld** in the north, and the more temperate high veld in the south.

WORLD CLIMATIC TYPES

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

DISTRIBUTION



10 The word veld is a Dutch word given by the early pioneer Dutch farmers who came to settle here.

11 It means “field” and is pronounced as “felt”.

12 In Australia, the grasslands are better known as Downs and are found in Murray-Darling basin of Southern Australia.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

a

TEMPERATURE

- 1 Their location in the heart of continents means that they have little maritime influence.
- 2 Their Climate is thus continental with extremes of temperature.
- 3 Summers are very warm over 19°C .
- 4 Winters are very cold in the continental steppes of Eurasia because of the enormous distances from the nearest sea.

- 5 The winter months are well below freezing.
- 6 In contrast, the Steppes type of climate in the southern hemisphere is never severe.
- 7 The winters are so mild that the mean temperature for any of the winter months is usually between $2-12^{\circ}\text{C}$.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

a

TEMPERATURE



WINTER IN EURASIAN STEPPES

8

The annual range of temperature is great, a direct result of continentality.

9

Winters are so cold that parts of the Eurasian Steppes are snow-covered for several months.

10

The snow melts with the return of spring and by mid-summer, temperatures soar to over 19°C . It is really hot for its latitude.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

b

PRECIPITATION

- 1 In its continental position, the annual precipitation of the Steppe Climate can be expected to be light.
- 2 The average rainfall may be taken as about 50 cm, but this again varies according to location from 25-75cm.
- 3 Most of the winter months have about an inch of precipitation, brought by the occasional Westerlies and coming in the form of snow.
- 4 In many other continental stations, the annual precipitation is even less, though the general pattern remains the same with most of the rain falling in the summer.
- 5 The maritime influence in the Steppe type of Climate of the Southern Hemisphere is even better brought out by the rainfall regime.

5

THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

b

PRECIPITATION

6

Its annual precipitation is always more than the average 20 inches because of the ocean present.

7

For example, Pretoria has an annual precipitation of 66 cm with the wettest months in November, December, January and February, the summer season of the southern hemisphere.

8

There are three months without any rain and this is the period of drought that may have such a disastrous effect on the sheep rearing industry here.

9

The dry season is particularly pronounced in temperate grasslands adjoining deserts, for example in Australia.

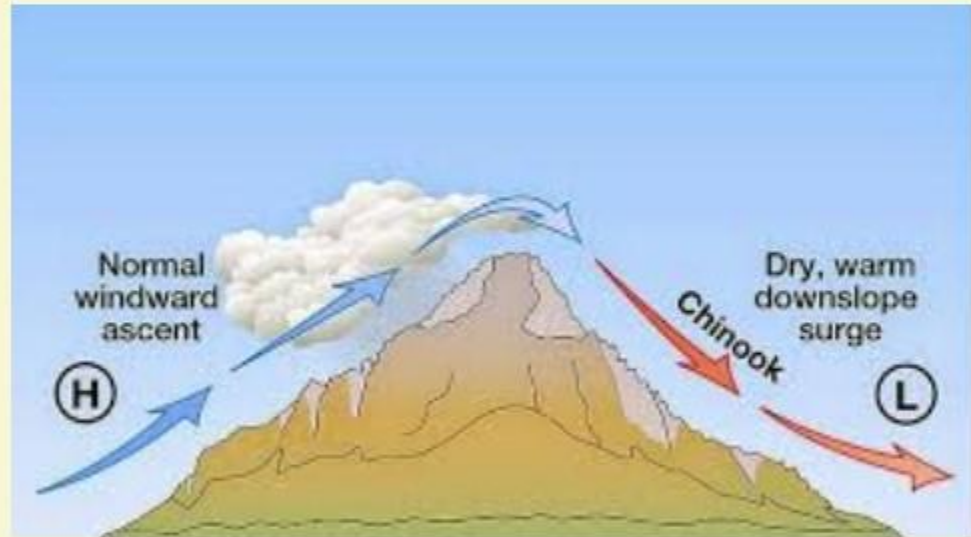
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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

b

PRECIPITATION



CHINOOK

10

On the Eastern slopes of the Rockies in Canada and U.S.A. a local wind, similar to the Fohn in Switzerland, called the Chinook, comes on to the Prairies and has a considerably effect on the local pastures.

11

It actually comes with the depressions in winter or early spring from the Pacific coast ascending the Rockies and then descending to the Prairies.

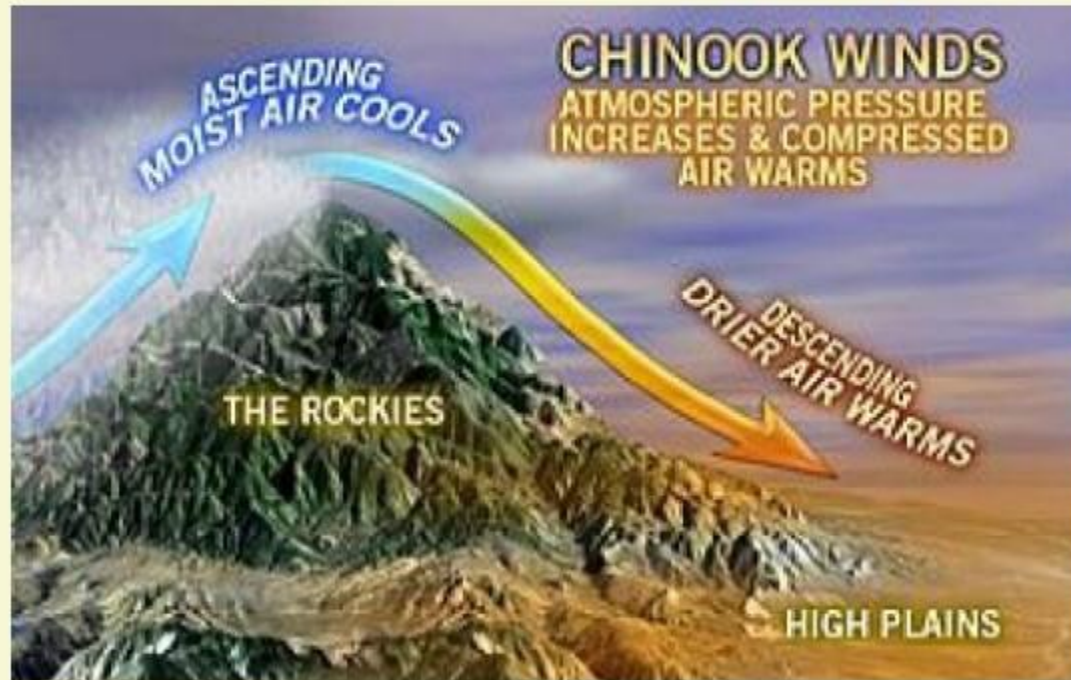
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THE TEMPERATE CONTINENTAL (STEPPE) CLIMATE

CLIMATE

b

PRECIPITATION



CHINOOK

12

It is a hot wind and may raise the temperature by 4°C within a matter of minutes.

13

It melts the snow-covered pastures and animals can be driven out of doors to graze in the open fields.

14

The agricultural year is thus accelerated. Local farmers welcome the Chinook for frequent Chinooks mean mild winters.

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